

SUMMARY

Software Engineer with 3+ years of experience developing healthcare and retail systems using Java, Spring Boot, Hibernate, Python, and Node.js for backend services, alongside React and Angular for frontend development. Proficient in microservices, REST and GraphQL APIs, event-driven architectures with Kafka and RabbitMQ, and cloud-native solutions on AWS leveraging Docker, Kubernetes, and CI/CD pipelines. Experienced in deploying ML-powered analytics, anomaly detection, and AI-driven RAG retrieval workflows with Hugging Face Transformers, LangChain, and vector databases to streamline operations, reporting, and internal processes.

EXPERIENCE

CVS Health, USA | Software Engineer

July 2025 – Present

- Architected scalable backend microservices using Java 17, Spring Boot, Hibernate, and PostgreSQL to handle patient claims, prescription fulfillment, internal operations workflows, processing 50,000+ daily transactions, reducing system latency from 6s to 2s per request.
- Built interactive dashboards and tooling using React.js, TypeScript, Material UI, consolidating patient, pharmacy, claims data streams, enabling 200+ internal users to review operational metrics, compliance data in near real-time.
- Constructed AI-powered document retrieval, summarization workflows using Hugging Face Transformers, LangChain, Pinecone vector databases, enabling operations teams to access policy, regulatory, runbook documents, cutting research time by 70%.
- Designed event-driven, cloud-native pipelines using AWS Lambda, SQS, Step Functions, Docker, orchestrating prescription, claims processing tasks with automated retries, alerting, improving processing throughput by 2X.
- Implemented data validation, logging, and error-handling mechanisms across microservices and AI workflows, ensuring data integrity, regulatory compliance, and traceability for 100% of patient and claims transactions.
- Led CI/CD, observability enhancements using GitHub Actions, Docker, Kubernetes (EKS), Prometheus, Grafana, achieving 99.95% service uptime, reducing mean time to resolution for incidents during high-demand pharmacy cycles.

KPIT, India | Software Engineer

April 2021 – January 2024

- Engineered backend services for retail transaction and inventory management using Java, Spring Boot, Spring MVC, Hibernate, and REST APIs, processing 15,000+ daily transactions and reducing order processing latency from 5s to under 2s.
- Migrated core databases from MySQL to PostgreSQL with optimized indexing, partitioning, and query strategies, reducing batch reconciliation runtime by 35 minutes per cycle and increasing reporting efficiency.
- Applied Redis caching and Elasticsearch-based search for product catalog and inventory APIs, reducing API response time from 400ms to under 120ms and improving end-user experience for 50,000+ catalog requests daily.
- Formed interactive React.js dashboards with TypeScript and Express.js APIs, enhancing visibility for 200+ retail operations users, enabling real-time stock monitoring, and reducing manual issue triage time by 50%.
- Developed event-driven microservices using Apache Kafka and RabbitMQ to decouple inventory, billing, and order workflows, improving system throughput by 2x and minimizing downtime during peak retail operations.
- Integrated AWS cloud-native infrastructure (EC2, S3, Lambda, RDS) with Docker and Kubernetes orchestration, reducing deployment setup time from days to 3 hours and scaling efficiently during high-volume retail events.
- Monitored application health and performance using Prometheus and Grafana, setting up alerting for CPU, memory, and transaction anomalies, enabling resolution of issues and minimizing downtime during retail sales spikes.
- Orchestrated automated testing frameworks using JUnit, Mockito, and Postman for API and workflow validation, reducing post-release defect resolution time by 30% and ensuring quality compliance.
- Established CI/CD pipelines with GitHub Actions, Jenkins, and Terraform, standardizing builds, tests, and deployments, enabling consistent production releases with minimal manual intervention.

KEY PROJECTS

AI-Powered Customer Support Chatbot

JavaScript, Node.js, Express, MongoDB, Hugging Face Transformers, RAG pipelines, Pinecone, Docker

- Formulated a full-stack AI chatbot using Node.js backend and React frontend, integrating RAG pipelines with Hugging Face and Pinecone to provide semantic responses, reducing simulated query resolution time from 8 minutes to under 2 minutes across 5,000+ mock interactions.
- Deployed Dockerized services on AWS ECS with automated CI/CD using GitHub Actions, enabling 20+ production updates per month with near-zero downtime and improved reliability for internal testing teams.

Predictive Sales & Recommendation Engine

React, Node.js, Express, MySQL, Python, Scikit-learn, Kafka

- Devised a recommendation engine and predictive sales model using Python ML pipelines and Node.js APIs, generating personalized product suggestions for 50,000+ simulated users and improving click-through insights for targeted campaigns.
- Created interactive React dashboards to visualize predicted sales and customer behavior, integrating Kafka streams for real-time updates, reducing manual reporting effort by 10+ hours per week for the analytics team

SKILLS

Programming & Scripting Languages: Python, Java, JavaScript, TypeScript, SQL, C++**Frontend & UI Development:** React, React Native, Next.js, Vue.js, Redux Toolkit, Tailwind CSS, Material UI, Mantine, Shadcn/UI**Backend & Service Frameworks:** Node.js, Express.js, Spring Boot, Flask, GraphQL, gRPC, tRPC**Databases & Storage Solutions:** PostgreSQL, MySQL, MongoDB, Redis, Cassandra, DynamoDB, Firebase, OracleDB**Artificial Intelligence & LLMs:** LangChain, RAG pipelines, OpenAI API, Vector Databases, Weaviate, Embedding-based Search, Prompt Engineering**Cloud Infrastructure & DevOps Tools:** AWS (EC2, S3, Lambda), RDS, Docker, Kubernetes, CloudWatch, GitHub Actions, GitLab, Jenkins, Kibana**Testing & QA Tools:** Jest, Mocha, Cypress, JUnit, Postman, PyTest**Methodologies & Practices:** Agile/Scrum workflows, Microservices Architecture, Event-driven Systems, CI/CD Automation, DevOps Best Practices

EDUCATION

Master of Science in Computer Science | **University at Buffalo –The State University of New York**

Buffalo, NY, USA

Bachelor of Technology in Computer Science | **SRKR Engineering College, Bhimavaram**

Andhra Pradesh, India